Small-Object Detection in Water Bodies Using the Data Captured by USV

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***Abstract*— The contamination of natural water bodies with small plastic water bottles and cans poses a pressing environmental challenge. To address this issue, this project focuses on the development of an automated system for small plastic water bottles and cans detection within water bodies. In this project, we lay the foundation for this system by leveraging Unmanned Surface Vehicles (USVs) to capture high-resolution imagery of water bodies. The primary objective is to demonstrate the feasibility of using computer vision technology to detect and quantify the presence of small plastic bottles and cans in water bodies. Convolutional Neural Networks (CNNs) are highly effective for object detection tasks. Techniques like Faster R-CNN, YOLO (You Only Look Once), and SSD (Single Shot MultiBox Detector) have shown excellent performance in detecting objects within images. These networks can be trained on a labeled dataset of water bottles and cans to identify them accurately. Apart from Deep learning techniques, Image segmentation techniques can be used to separate objects of interest (water bottles and cans) from the background. In this project we will be working on Semantic Segmentation and Instance Segmentation. Semantic Segmentation is a technique which assigns each pixel in the image to a class (e.g., "water bottle," "can," or "background"), creating a pixel-wise mask of the objects. And Instance Segmentation goes a step further and not only classifies objects but also distinguishes between individual instances of the same class. It can be useful when you want to count the number of bottles or cans.**

Keywords—YOLO, FloWImg, bottle, small object, hornet, bounding-box.

I. INTRODUCTION

Waste in water bodies is a modern problem that has grown to alarming proportions. It is a complex problem with wide-ranging environmental effects. Globally, water bodies—from lakes and rivers to the oceans—bear the brunt of human activity that fuels the unrelenting buildup of waste. Humans are dumping waste like bottles in water bodies directly as shown in Fig 1. Aside from degrading aquatic ecosystems' aesthetic appeal, plastics, chemicals, and other non-biodegradable materials also seriously endanger aquatic life and human health. Waste is ubiquitous in water bodies, upsetting delicate ecological balances and eventually causing the deterioration of entire ecosystems and the quality of the water to decline. In order to reduce the number of pollutants entering our essential water supplies, creative solutions and increased awareness are urgently needed, as shown by the current efforts to address this issue. It is becoming more and more clear that sustainable practises and technologies that support efficient waste management in water bodies are necessary as communities deal with the fallout from careless waste disposal.

The pressing need for innovative technology to tackle environmental deterioration and pollution has been highlighted in light of the growing concerns around these issues in recent times. It is a difficult but necessary duty to detect and monitor contaminants in water bodies, especially tiny items like waste products. Modern computer vision models have enabled Unmanned Aerial Vehicles (UAVs) to become extremely effective instruments for environmental surveillance. This study uses data from unmanned aerial vehicles (UAVs) to identify tiny things in water bodies using YOLO (You Only Look Once) object detection algorithms, especially versions V5, V7, and V8.

The YOLO framework, which is well-known for its ability to recognize objects in real time, has undergone several iterations of improvement that have increased the model's precision and effectiveness. This study investigates the ability of YOLO versions V5, V7, and V8 to identify and categorize tiny items in water bodies, with a major focus on garbage. The data collected by UAVs is an invaluable asset that provides an aerial perspective that enables thorough observation of aquatic habitats.

A crucial component of any detection system, the accuracy metrics are essential for assessing how well the models that have been put into practice are working. Our research yields encouraging findings: YOLO V5 detects tiny things in water bodies as depicted in Fig 2, with an accuracy of 34%, V7 with a 76% accuracy, and V8 with a remarkable 77.4% accuracy. These results highlight the YOLO versions' developing capabilities as well as the potential of unmanned aerial vehicles (UAVs) to transform environmental monitoring and lessen water pollution.

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Fig. 1. Sample image of waste floating on water

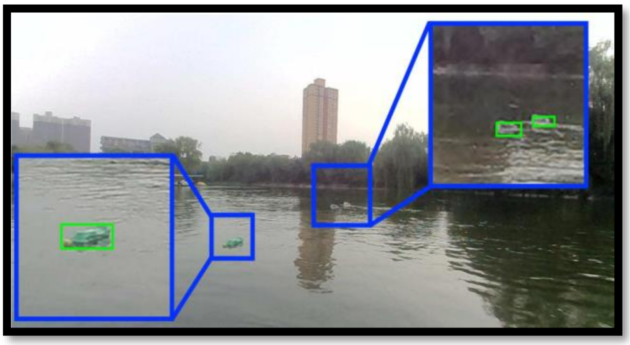

Fig. 2. Detection of waste floating on water

II. Litrature Survey

Water pollution, particularly from non-biodegradable materials like plastics and chemicals, poses a significant threat to aquatic ecosystems. One effective way to mitigate this issue is by intercepting and removing these wastes in inland waters before they reach the ocean. This study focuses on detecting water bottles and cans in such environments, inspired by similar methods used in previous research [1,10].

AquaVision, a 2020 study, pioneered the use of deep transfer learning for waste detection in water bodies [3]. This study utilized Convolutional Neural Networks (CNNs) like ResNet, VGG, and MobileNet, which were pre-trained on large datasets to identify waste patterns in diverse water environments. Following AquaVision, 2021 saw various innovative approaches to small object detection in water bodies. One approach involved using wireless sensor signals to detect floating debris based on changes in frequency around the waste [4]. Another method used high-resolution multispectral images from the Sentinel-2 satellite to detect floating objects in coastal areas [5]. The MacroFloat dataset from this study has significantly contributed to research on marine pollution and coastal safety.

In 2021, another innovative approach used millimeter-wave radar to obtain synchronized images for waste detection [6,10]. This radar technology operates in the 30 to 300 GHz frequency range, offering detailed imaging capabilities.

In 2022, a study utilized drone images and deep learning techniques to identify temporary water bodies in real time [2]. This research provided insights into working with data collected by unmanned surface vehicles (USVs), similar to the unmanned aerial vehicles (UAVs) used in their study. Additionally, research on infrared images demonstrated advanced methods for detecting small objects in water surfaces, even under adverse weather conditions [7]. This model used spatial feature weighting and class balancing to enhance detection accuracy for small objects with low thermal signatures.

Recent studies in 2023 have further advanced the field. One notable publication utilized YOLOv5 with Soft-NMS and SIOU for detecting small water surface floaters in UAV-captured images [8]. YOLOv5, a real-time object detection system, predicts bounding boxes and class probabilities for each grid cell in an image. Soft-NMS improves bounding box predictions by adjusting scores for overlapping boxes, while SIOU evaluates the precision of detected regions.

Another significant study analyzed deep learning models like De-Nosing Auto Encoding (DAE) and Long Short-Term Memory (LSTM) for garbage detection in water bodies [1]. These models have shown promising results in accurately detecting waste in aquatic environments.

Our study builds on these advancements by using a USV-based dataset and applying YOLO models with enhanced activation functions such as Leaky ReLU, Hardswish, and SiLU. Additionally, we incorporated transformers like SETR, DETR, and C3tr to improve detection accuracy. These improvements aim to significantly enhance the system's performance in detecting small objects, which is crucial for pollution management and environmental monitoring.

In summary, our research aims to develop a robust system for detecting small objects in water bodies using data captured by USVs. By leveraging advanced deep learning techniques and enhancing model architectures, we strive to achieve higher detection accuracy and contribute to more effective environmental protection efforts.

III. Methodology

A. *Dataset:*

The FloW-IMG dataset from ORCA-Uboat, used in our study [3], includes 2,000 .jpg images with over 5,000 labeled instances of floating garbage, primarily plastic bottles. The dataset, captured by an unmanned surface vehicle (USV) in inland water bodies like lakes, features images at resolutions of 1280x640 and 1280x720. Objects of interest are small, not exceeding 32x32 pixels.

Figure 3 illustrates various detection challenges within the dataset. Figures 3A and 3B show images with brightness and contrast issues. Figures 3C and 3D depict clearly visible bottles and cans. Figures 3E and 3F show objects at inappropriate angles, while Figures 3G and 3H lack bottles or cans. Figures 3I and 3J are blurred or unclear, making detection difficult.

Brightness or Contrast Faulty

Fig. 3A

Fig. 3B

Clear Visibility

Fig. 3C

Fig. 3D

Fig. 3. Challenges faced with the FloWImg Dataset.

We also incorporated water bottle images from different sources, categorized and shown in Figure 4.2. The training, validation, and testing sets contain 7,800, 1,800, and 184 images respectively, labeled as broken, normal water, normal cap, no cap, foreign object, and scratched. We applied image preprocessing and data augmentation.

This dataset enhanced our model's ability to detect plastic waste in water. It includes 472 water bottle images classified into Half water level, Full water level, and Overflowing.

B. Data Preprocessing:

The FloW-IMG dataset, provided by Chinese USV manufacturer ORCA-Uboat, was used in this project. It features a substantial collection of floating bottle images annotated with PASCAL Visual Object Classes (VOC) XML format. As shown in Figure 6, the XML annotations include details such as image resolution, RGB channel depth, and bounding box coordinates. The class name "bottles" appears in the annotations, focusing solely on detection and localization rather than classification or segmentation. The truncated tag indicates whether the object is fully or partially visible, while the pose tag remains undefined due to varying orientations.

The chosen model for small object detection is YOLOv7, which requires annotations in a specific text format. However, the dataset from ORCA-Uboat only includes XML annotations. Therefore, the first step was to convert these XML annotations into the YOLO format. A simple Python function was developed to parse the XML files and generate the necessary text files for YOLO.

Understanding the YOLO annotation format is crucial for this conversion. Each line in the YOLO format text file represents an object, including its class, normalized bounding box dimensions, and the X and Y coordinates of its center point. All parameters are normalized to fit within the range of 0 to 1.

The FloW-IMG dataset, consisting of 2000 images, was split into 60% for training (1200 images) and 40% for validation and testing (800 images). The dataset presents challenges like glare, blurriness, and varying image quality. Data augmentation was applied to the training set, adjusting saturation, brightness, and exposure, expanding it to 3600 images to simulate real-world conditions. The YOLOv7 model, chosen for detection, uses a composite loss function including objectness, location, and box loss, with class loss set to zero due to the single object class. The system was evaluated using the 800-image test set.

C. Model Architecture:

Three essential parts make up the YOLOv7 architecture, which is well-known for its skill at object detection: the head, neck, and backbone. Together, these components enable remarkably accurate feature extraction, bounding box prediction, and object classification. Let's examine each part in more detail:

1. Head:

YOLOv7 utilizes multiple heads for different tasks:

Object detection head: This predicts bounding boxes and class probabilities for each detected object.

Confidence head: This predicts the confidence score for each bounding box, indicating how certain the model is about the detection.

Classification head: This further refines the class prediction for each object.

Focus mechanism: YOLOv7 employs a novel focus mechanism that dynamically adjusts object detection based on image content. This improves efficiency and accuracy, especially for small objects.

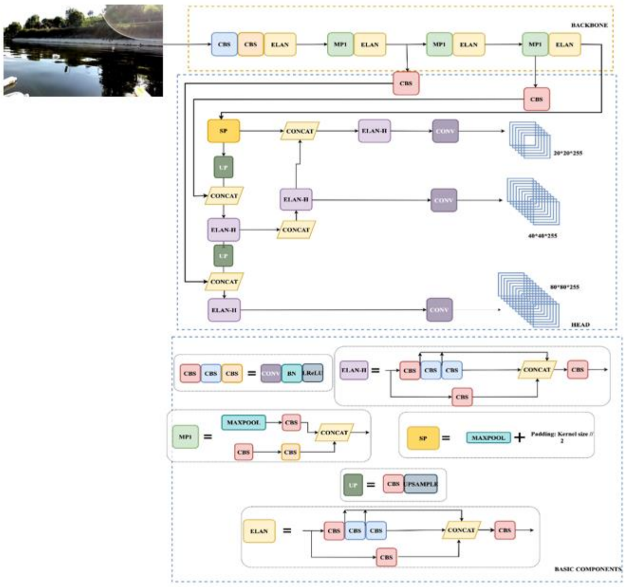

Fig. 4. The architecture of yolov7 model.

2. Neck:

The neck acts as a bridge between the backbone and head: It processes and refines the feature maps extracted from the backbone, preparing them for the head's prediction tasks.

YOLOv7 uses a Cross Stage Partial Connection (CSP) neck: This design reduces redundancy and improves information flow, leading to better performance compared to previous YOLO versions.

3. Backbone:

The backbone is the foundation of the network: It extracts features from the input image at different scales and resolutions.

YOLOv7 offers backbone flexibility: You can choose from various pre-trained backbones like CSPDarknet53, EfficientNet, and ConvNeXt, depending on your desired accuracy and inference speed trade-off.

D. Proposed Model Architecture:

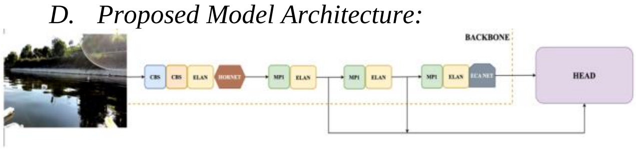

Fig. 5. Proposed model architecture after fine tuning the model.

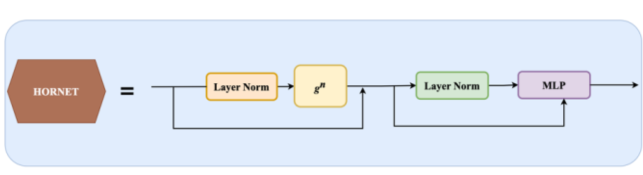

Fig. 6. HORNET architecture.

Although the YOLOv7 architecture in its basic form has remarkable object detection capabilities, researchers are always looking for ways to improve its effectiveness. To increase feature extraction and accuracy, one such investigation is adding Hornet to the backbone.

1. Hornet:

Channel-wise transformers: Hornet replaces traditional convolutional layers with lightweight transformers operating on individual channels. This allows for better long-range feature interactions and dependencies, leading to higher-quality feature maps.

Efficient computation: Hornet utilizes a simplified encoder-decoder structure and dynamic skip connections, resulting in lower computational cost compared to standard transformer architectures. This efficiency makes it suitable for real-time object detection on resource-constrained devices.

The convolutional layers of the conventional YOLOv7 architecture use the ReLU activation function. Although successful, scientists are looking into different activations that can enhance the network's functionality. Here is a brief summary of the applications of SiLU, Leaky ReLU, and Parametric ReLU in modified YOLOv7 architectures.

E. Training:

Preparing the YOLO standard annotations for the dataset of floating bottles in inland water bodies was the initial stage in our approach. In this approach, a little item was defined as one that covered an image with a maximum of 32 by 32 pixels. The YOLOv7 model was selected since it was pre-trained on the COCO dataset and has better accuracy than its predecessors. Using our dataset, the pre-trained model is refined through the application of transfer learning. 2000 photos and comments made up the dataset, which was divided into training and testing sets of three to two.

The goal was to identify and locate bottles, as the Flow dataset contained only one type of object: bottles. Pretrained weights acquired during training on the COCO dataset were used to initialise the model. The architecture was created so that it could be trained using 640 x 640 pixel input images. Table 1 lists the model's training parameters. As seen in Table 2, we applied a number of data augmentation methods, such as HSV, translation, scaling, horizontal flipping, and glaring, to further enhance the performance of our model. Hue, saturation, and value shifts were among the range of values that were applied to the HSV augmentation. Random applications were made to the translation, scaling, and horizontal flipping augmentations. To make the training more diverse, mosaic augmentation was applied over the entire set of training data. Google Colab Pro was used for the training process, and the GPU accelerator was used to speed up the process. The validation dataset was used to assess the model during training to make sure it wasn't overfitting to the training dataset.

Figure 7 in the documentation showcases the plots of both training and validation losses. The observed decrease in these losses suggests that the model is learning effectively during training and is generalizing well to the validation data. This decrease in losses is a positive indication of the model's performance.

TABLE I. HYPERPARAMETER USED IN MODEL TRAINING

Hyperparameter	**Value**
Learning Rate	0.01
Momentum	0.937
Weight Decay	0.0001 to 0.001.
Optimizer	Stochastic Gradient Descent
Epochs	100
Batch Size	16
Image Size	640
Dropout	0.3

To comprehensively evaluate the model, three key performance metrics were considered: recall, precision, and mean average precision (mAP). Recall measures the model's ability to identify all relevant instances within the data, precision assesses the accuracy of the positive predictions made by the model, and mAP provides an overall measure of the model's detection performance at different intersection over union (IoU) thresholds. Specifically, mAP 0.5 evaluates the average precision at an IoU threshold of 0.5, while mAP 0.5:0.95 averages the precision across multiple IoU thresholds ranging from 0.5 to 0.95.

The results, visualized using tools such as Comet and ClearML, demonstrate improvements in these metrics, indicating the model's increasing accuracy and reliability. This performance is significant as it suggests that the model can effectively detect floating bottles in various challenging conditions, making it suitable for real-world applications.

In summary, the data augmentation strategy significantly improved the training set, enabling the model to perform well under diverse conditions. The comprehensive evaluation metrics and visualizations underscore the model's robust performance and its potential for effective deployment in detecting floating bottles across different scenarios..

IV. RESULT

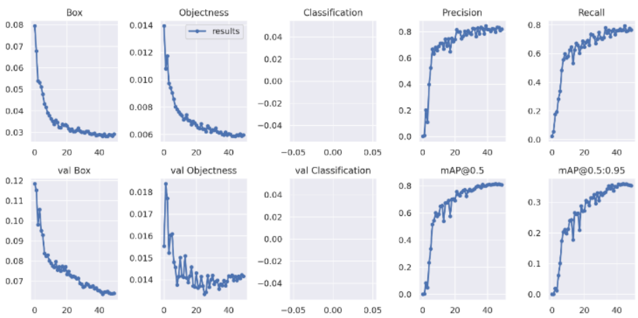

Fig. 7. Performance metrics of the proposed model.

The graphs in Figure 7 demonstrate effective model training, indicated by decreasing training box loss, class loss, and DFL loss. This reduction signifies that the model's predictions are becoming increasingly accurate compared to the actual target values in the training set. Additionally, the gradual decrease in validation loss indicates that the model is generalizing well to new, unseen data, avoiding overfitting. Monitoring validation loss is crucial for ensuring ongoing model performance on unfamiliar data and may guide hyperparameter tuning and model selection.

Precision and recall metrics are also improving, reflecting the model's enhanced accuracy in making positive predictions (precision) and correctly identifying a larger portion of actual positive instances (recall). This suggests effective model refinements and threshold adjustments. For floating bottle detection, our proposed model achieved a mean Average Precision (mAP) score of 84.7%, a 37.5% improvement over the baseline models referenced in [3]. Table 3 highlights the superior performance of our model against these baselines for the FloW-IMG dataset.

Table II illustrates the enhanced performance metrics of the modified YOLOv7 model, especially with the integration of the Hornet module and PreLU (Parametric ReLU) activation function. This model achieved an impressive mAP of 84.7% and an mAP (0.5:0.95) of 39.5%, showcasing its robustness and accuracy in bottle detection. The integration of the Hornet module with ECAnet and the PreLU activation function significantly improves the model's ability to capture complex features, leading to better generalization and performance on the test set. The mAP score of 84.7% indicates a high level of precision across different detection thresholds, while the mAP (0.5:0.95) score of 39.5% underscores the model's versatility in maintaining performance across various IoU thresholds.

TABLE II. RESULTS OBTAINED AFTER TRAINING DIFFERENT MODELS

Models	**mAP 0.5/ %**	**mAP 0.5:0.95/ %**	**Precison/ %**	**Recall/ %**
YOLO v5	44.4	19	41.6	34.6
YOLO v7	61.3	23.24	46.4	69.6
YOLO-V7 -Tiny Hornet	63.4	25.9	73	59.8
YOLO-V7-hornet-ecanet-Pipe	79.1	36.2	81.2	78.3
YOLOv7 -tiny-ecanet with new dataset	77.1	31.2	83	71.6
YOLOv7 Tiny-ecanet with new bottle dataset	83.1	36.4	81.5	76.8
YOLOv7 -tiny-hornet-ecanet- pipe with new bottle dataset	**84.7**	**39.5**	**83.9**	**81.2**

In Figure 15, test set images reflect the model's performance. High light reflections on the water's surface in images 15.a challenge the model, but it still predicts accurately, albeit with low confidence due to reduced clarity. Images 15.b display bottles at various complexities. The model correctly detects bottles in 15.b where they are clear and well-separated. Despite some overlap in 15.b, the model distinguishes five different bottles, though with slightly lower confidence scores.

Fig. 15A

Fig. 15B

Fig. 15C

Fig. 15D

Fig. 8. Analyzing model predictions on test set.

Images 15.c and 15.f present camouflage scenarios where the model struggles to detect bottles, as they are well-hidden against the background. Image 15.d shows correct identification despite the small target size, but also includes some false predictions due to insufficient distinguishing information. Occlusion in 15.j affects the model's ability to gather adequate context, resulting in poor performance.

Overall, the model proves robust against strong light reflections and capable of identifying bottles under varied complexities when clearly visible. However, it faces difficulties with occlusion, camouflage, and insufficient data, highlighting areas for further improvement.

V. CONCLUSION

We evaluate several conventional vision-based object detection algorithms and radar detectors on our dataset. The results of the experiment show that in order to meet the requirements of real-world applications, the robustness of the model for vision-based object detection must be strengthened. We anticipate that the cleanup of wastes in water regions will receive more attention, and that detection of floating wastes may become more accurate. In some unclean areas, we will collect more types of floating trash for more categories in the future to encourage the use of autonomous cleaning in inland waterways. As we have already trained with the different versions of the yolo with small, medium and large size object detections and compared with the mAP's of each model. The future scope of the project is to modify the architecture of the yolo versions and add attention mechanism, bounding box algorithm and the modify the backbone architecture so that the mAP of the proposed model will give the satisfactory results.

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